WORKED SOLUTION

STEP II 1995, Question 3

Pure • Logical Deduction and Counting

> **STEP II 1995, Question 3**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/95-s2-q3

The key idea

This is a logic puzzle in disguise, and the way in is *counting*. Every stage hands out $5+4+3+2+1=15$ points, so the five scores add up to a known total. Arouet's score is so close to the maximum that it pins down nearly all of his results. After that, Eluard's results squeeze Eluard's score from both sides, and Camus's "same position four times" leaves only a few cases to check.

Solution

We write A, B, C, D, E for the riders' total scores. We're told $A > B > C > D > E$ and $A = 24$.

The total, and Arouet's stages

Each stage gives out 15 points, so over five stages

$$A+B+C+D+E=75, \qquad \text{hence} \qquad B+C+D+E=51.$$

Arouet can get at most $5\times 5=25$ points. Getting 24 means losing exactly one point, so Arouet came *first in four stages and second in one*. Eluard won stage 5, so Arouet's second place must have come in stage 5, and Arouet won stages 1–4.

Why Eluard scored 11

Lower bound. Eluard got 3 points in stage 4 and 5 in stage 5, and at least 1 point in each of the other three stages. So $E \geq 3+5+3=11$.

Upper bound. The scores are distinct integers with $B > C > D > E$, so $D \geq E+1$, $C \geq E+2$ and $B \geq E+3$. Therefore

$$51 = B+C+D+E \geq 4E+6, \qquad \text{so} \qquad E \leq \tfrac{45}{4} < 12.$$

Combining the two bounds gives $E=11$. Equality in the lower bound also tells us that

Eluard came *last* in each of stages 1, 2 and 3.

What is left for Barthes, Camus and Diderot

We now know where Arouet and Eluard finished in every stage, so Barthes, Camus and Diderot shared out the remaining places:

Stage	Arouet, Eluard	Places left for B, C, D	
1, 2, 3	1st, 5th	2nd, 3rd, 4th	(4, 3, 2 points)
4	1st, 3rd	2nd, 4th, 5th	(4, 2, 1 points)
5	2nd, 1st	3rd, 4th, 5th	(3, 2, 1 points)

Also $B + C + D = 51 - 11 = 40$, and $B > C > D > 11$, so $D \geq 12$ and $C \geq 13$.

Ask yourself Camus finished in the same position four times. Which positions could that be, given the table?

The answer is 2nd, 3rd or 4th. Fifth place was available to Camus only in stages 4 and 5, which is two stages, not four. We take the three possibilities in turn.

Camus 2nd four times. Second place was available only in stages 1–4, so Camus was second in all of them. Then in stages 1–3 Barthes got at most 3 points each, in stage 4 at most 2, and in stage 5 at most 3. So $B \leq 9 + 2 + 3 = 14$. But $C \geq 4 \times 4 = 16 > B$, which is a contradiction.

Camus 4th four times. Fourth place is worth 2 points in every stage, so Camus's best total is $4 \times 2 + 4 = 12$ (a second place in the remaining stage). That contradicts $C \geq 13$.

Camus 3rd four times. Third place was available in stages 1, 2, 3 and 5, worth 3 points each time, so these are Camus's four third places. In stage 4 Camus scored $x \in \{4, 2, 1\}$, and $C = 12 + x$.

- $x = 4$: $C = 16$, so $B \geq 17$ and $D = 40 - B - C \leq 7$. This contradicts $D \geq 12$.
- $x = 2$: $C = 14$, so $B \geq 15$ and $D \leq 11$. Again a contradiction.
- $x = 1$: $C = 13$, so $B + D = 27$ with $B \geq 14$ and $12 \leq D \leq 12$. Hence $D = 12$ and $B = 15$.

So the scores are

$$\boxed{B = 15, \quad C = 13, \quad D = 12} \qquad (\text{with } A = 24,\ E = 11).$$

Barthes in the final stage

Ask yourself We know the totals, but not the stage-by-stage results. Is there enough information to fix stage 5?

There is, by a parity argument. With Camus third in stages 1, 2, 3 and 5 and fifth in stage 4, Barthes and Diderot shared the places worth 4 and 2 in each of stages 1–4. In

stage 5 they shared the places worth 2 and 1 (4th and 5th). Suppose Diderot took the 4-point place in k of stages 1–4 and scored $s \in \{1, 2\}$ in stage 5. Then

$$D = 4 \times 2 + 2k + s = 8 + 2k + s = 12.$$

So $s = 4 - 2k$ is even, which forces $s = 2$ (and $k = 1$). Diderot was fourth in the final stage, so **Barthes came last (fifth) in the final stage.**

Such a tour does exist. For example, take stage 1: A, D, C, B, E; stages 2 and 3: A, B, C, D, E; stage 4: A, B, E, D, C; stage 5: E, A, C, D, B. This gives $24, 15, 13, 12, 11$, as required.

Checks

- The totals add up: $24 + 15 + 13 + 12 + 11 = 75 = 5 \times 15$.
- In the example tour, Camus is third in four stages, and Eluard is third in stage 4 and first in stage 5.
- A computer search through all 120^5 possible sets of stage results, keeping those that meet every condition, finds only the scores $(24, 15, 13, 12, 11)$. Barthes is fifth in stage 5 in every case. (There are exactly four valid tours, depending on which of stages 1–4 Diderot finished second in.)

Common slips

- Showing only $E \geq 11$. The question asks why Eluard *must* have 11, so the upper bound from $B + C + D + E = 51$ and distinctness is essential.
- Assuming Camus was in the same place in four *consecutive* stages, or forgetting the case of fourth place.
- Finding the scores but not proving that the final-stage placing is forced. One example tour isn't enough; the parity argument shows that every possible tour has Barthes last in stage 5.

Beyond STEP

The argument has three moves: a conservation law (the fixed total), extremal reasoning (24 is nearly the maximum), and a parity argument. All three appear again and again in combinatorics. Bounding a quantity from both sides until it is forced is also the basic technique of analysis and of olympiad problem solving. For more puzzles in this spirit, try Siklos, *Advanced Problems in Mathematics*. Engel's *Problem-Solving Strategies* has whole chapters on the invariance and extremal principles.

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